From Differential Privacy to Concentrated Privacy

A source-guided derivation of definitions, bounds, and composition

Technical study notes

2026-10-23

This note develops pure ε-DP, approximate (ε, δ)-DP, mean-concentrated DP (CDP/mCDP), and zero-concentrated DP (zCDP). It follows the mathematical progression in Dwork–Rothblum–Vadhan (2010), Dwork–Rothblum (2016), and Bun–Steinke (2016). Every major formula is preceded by the question it answers. Elementary derivations are given in full, including the algebra and the probabilistic lemmas they rest on; non-elementary results are named and cited, not silently assumed.

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1 The common mathematical object: two output distributions

Take neighbouring datasets $x \sim x'$ that differ in one person's record, and one randomized mechanism M. Write $P = \mathcal{L}(M(x))$ and $Q = \mathcal{L}(M(x'))$ for its two output distributions, which are probability measures. For an event S, $P(S)$ and $Q(S)$ are probabilities. If the two measures have densities p, q with respect to a common dominating measure ν, then

$$P(S) = \int_S p(y)\,\mathrm{d}\nu(y), \qquad Q(S) = \int_S q(y)\,\mathrm{d}\nu(y).$$

The measure ν is Lebesgue measure in continuous examples and counting measure in discrete ones (one can always take $\nu = P + Q$). The notation therefore never mistakes a density value for the probability of an individual output.

Whenever $P \ll Q$, define the likelihood ratio $R = \mathrm{d}P/\mathrm{d}Q$ and the **privacy loss** $L(y) = \log R(y)$. It is defined Q-almost surely; when we sample $Y \sim P$, $L(Y)$ is a random variable. Under pure DP, P and Q are mutually absolutely continuous, so the ratio is finite and bounded almost surely. Under approximate DP this need not be true: P may put mass where Q has none, and there we set $L = +\infty$.

Why almost surely? Suppose $P(S) \le e^{\varepsilon} Q(S)$ holds for every measurable event. Then $p(y) \le e^{\varepsilon} q(y)$ for ν-almost every y, though not necessarily at every point of an arbitrarily chosen version of the densities. The argument runs in both directions:

- If $p \le e^{\varepsilon} q$ holds ν-almost everywhere, integrating over S gives $P(S) \le e^{\varepsilon} Q(S)$ exactly, because a null set contributes nothing to an integral.
- If $p > e^{\varepsilon} q$ on a set B with $\nu(B) > 0$, then $\int_B (p - e^{\varepsilon} q)\,\mathrm{d}\nu > 0$ (a strictly positive function integrated over a set of positive measure), so $P(B) > e^{\varepsilon} Q(B)$ and the event $S = B$ breaks the inequality.

Whenever a density-ratio bound is used below, its scope is stated.

2 Pure ε-differential privacy

2.1 Definition and interpretation

The mechanism is ε-DP ($\varepsilon \geq 0$) when, for every ordered pair $x \sim x'$ and every measurable S,

$$P(S) \leq e^{\varepsilon} Q(S). \tag{2.1}$$

Since the neighbour relation is symmetric, the definition also applies to (x', x), so $Q(S) \leq e^{\varepsilon} P(S)$ holds too. By the argument of Section 1 applied in both orders, the equivalent likelihood-ratio condition is

$$e^{-\varepsilon} \leq R(y) \leq e^{\varepsilon} \quad Q\text{-a.s.}, \qquad -\varepsilon \leq L(y) \leq \varepsilon \quad P\text{-a.s.} \tag{2.2}$$

(The two statements agree because P and Q have the same null sets.) The guarantee bounds every event probability, not only typical events. At $\varepsilon = 0$, $P = Q$: the outputs are identically distributed on neighbouring inputs.

2.2 Why basic composition adds ε

What is being composed. An adaptive analyst runs mechanisms $M_1, \ldots, M_k$ in turn; the i-th may depend on the earlier outputs, $Y_i \sim M_i(x, y_{<i})$. The analyst's choices are the same function of the transcript whichever dataset is in use; only the data differ. The hypothesis is that for every fixed history $y_{<i}$ the mechanism $M_i(\cdot, y_{<i})$ is ε_i-DP. Write $p_i(\cdot \mid y_{<i})$ and $q_i(\cdot \mid y_{<i})$ for its output densities on x and x'.

The density of the whole transcript is the product of the conditional densities (the chain rule), on each dataset:

$$\frac{p(y_{1:k})}{q(y_{1:k})} = \prod_{i=1}^{k} \frac{p_i(y_i \mid y_{<i})}{q_i(y_i \mid y_{<i})}, \qquad \text{so} \qquad L_{1:k} = \sum_{i=1}^{k} L_i.$$

Each factor is a likelihood ratio of an ε_i-DP mechanism with its history fixed, so by (2.2) $|L_i| \leq \varepsilon_i$ almost surely, and hence $|L_{1:k}| \leq \sum_i \varepsilon_i$ almost surely. Integrating that density bound over an event (Section 1) gives the event inequality, so adaptive composition is $(\sum_i \varepsilon_i)$-DP. In particular, k identical guarantees give $k\varepsilon$-DP.

2.3 Post-processing

If g is any function applied to the output, then $\Pr[g(M(x)) \in S] = P(g^{-1}(S)) \leq e^{\varepsilon} Q(g^{-1}(S))$, because $g^{-1}(S)$ is itself an event. A randomized g is a mixture of deterministic ones, and averaging the inequality over the mixture preserves it. The same one-line argument works for (ε, δ)-DP.

2.4 Example: Laplace noise and sensitivity

For a numerical query f with sensitivity $\Delta = \sup_{x \sim x'} |f(x) - f(x')|$, release $Y = f(x) + Z$ with $Z \sim \text{Laplace}(0, b)$, whose density is $p_x(y) = (2b)^{-1} \exp(-|y - f(x)|/b)$. The normalizing constants cancel in the ratio, leaving

$$\log \frac{p_x(y)}{p_{x'}(y)} = \frac{|y - f(x')| - |y - f(x)|}{b} \leq \frac{|f(x) - f(x')|}{b} \leq \frac{\Delta}{b}.$$

The first inequality is the triangle inequality, $|u| - |v| \leq |u - v|$ with $u = y - f(x')$ and $v = y - f(x)$. Setting $b = \Delta/\varepsilon$ gives pure ε-DP. This calculation bounds the ratio at every point for this continuous version of the density, which is stronger than the almost-sure bound the definition needs. For a vector-valued f, use independent Laplace noise in each coordinate and the ℓ_1-sensitivity; the same calculation applies coordinate by coordinate and the bounds add.

3 Approximate (ε, δ)-differential privacy

3.1 Definition: what δ actually means

For $\varepsilon \geq 0$ and $0 \leq \delta \leq 1$, we require, for every neighbouring ordered pair and every event S,

$$P(S) \leq e^{\varepsilon} Q(S) + \delta. \tag{3.1}$$

Unlike pure DP, this does not give a likelihood ratio that is bounded almost surely. It allows an additive discrepancy in the probability of an event. It is **not** in general equivalent to "the privacy loss exceeds ε with probability at most δ" with the same parameters; Section 3.3 makes both directions precise.

3.2 The exact meaning of δ: the hockey-stick divergence

Define $H_{\varepsilon}(P \| Q) = \sup_S [P(S) - e^{\varepsilon} Q(S)]$. With densities, the supremum is achieved by the set where $p(y) > e^{\varepsilon} q(y)$, so

$$H_{\varepsilon}(P \| Q) = \int \left(p(y) - e^{\varepsilon} q(y)\right)_+ \mathrm{d}\nu(y), \qquad u_+ = \max(u, 0). \tag{3.2}$$

To see this, write the difference for an event as $\int_S (p - e^{\varepsilon} q)\, \mathrm{d}\nu$. Adding to S points where the integrand is negative can only lower it, and adding every point where it is positive raises it to its maximum. Hence (ε, δ)-DP is exactly $H_{\varepsilon}(P \| Q) \leq \delta$ **and** $H_{\varepsilon}(Q \| P) \leq \delta$ for every neighbouring pair. Both directions are needed because H_{ε} is not symmetric. This is the precise operational meaning of δ.

If P has mass that is singular with respect to Q (where $q = 0$), that mass counts in full towards H_{ε}. The loss can therefore be infinite on some outcomes while (ε, δ)-DP still holds.

3.3 From a tail bound on the loss to (ε, δ)-DP, and why the converse fails

Every conversion below ends with the same step, so we prove it once.

Lemma 3.1 (tail bound implies approximate DP). If $\Pr_{Y\sim P}[L(Y) > \varepsilon] \le \eta$, then $P(S) \le e^{\varepsilon} Q(S) + \eta$ for every event S.

Proof. Split S into $S_{\le} = S \cap \{L \le \varepsilon\}$ and $S_{>} = S \cap \{L > \varepsilon\}$. On $S_{\le}$ the density bound $p \le e^{\varepsilon} q$ holds (on the part where $q = 0$, $L = +\infty$, so that part lies in $S_{>}$), and integrating it gives $P(S_{\le}) \le e^{\varepsilon} Q(S_{\le}) \le e^{\varepsilon} Q(S)$. The other part has $P(S_{>}) \le P(L > \varepsilon) \le \eta$. Add the two. □

The same computation gives an exact formula. On the optimal set $\{L > \varepsilon\}$, $q = e^{-L} p$, so

$$H_{\varepsilon}(P\|Q) = \mathbb{E}_P[(1 - e^{\varepsilon - L})_+] \le \Pr_P[L > \varepsilon]. \tag{3.3}$$

The weight $1 - e^{\varepsilon - L}$ is close to 0 when L only slightly exceeds ε and close to 1 when L is far above it. So δ is the tail probability of the loss *discounted* by how far the loss goes beyond ε.

The converse fails. Take $\varepsilon = 0$, $P = (\frac{1}{2}, \frac{1}{2})$ and $Q = (\frac{1}{2} + \delta, \frac{1}{2} - \delta)$ on two outcomes. Then $H_0(P\|Q) = H_0(Q\|P) = \delta$, so the pair is $(0, \delta)$-indistinguishable. Yet $L > 0$ on the second outcome, which has probability $\frac{1}{2}$ under P, however small δ is.

3.4 Two probabilistic tools

The concentration arguments below use only these two facts.

Chernoff's bound. For any random variable X, any t and any $\lambda > 0$, Markov's inequality applied to the nonnegative variable $e^{\lambda X}$ gives

$$\Pr[X \ge t] = \Pr[e^{\lambda X} \ge e^{\lambda t}] \le e^{-\lambda t}\, \mathbb{E}[e^{\lambda X}]. \tag{3.4}$$

Hoeffding's lemma (cited, not proved). If $\mathbb{E}X = 0$ and $a \le X \le b$ almost surely, then $\mathbb{E}[e^{\lambda X}] \le \exp(\lambda^2 (b-a)^2/8)$ for every real λ. The same holds conditionally on a σ-field when the conditional mean is 0 and the range has width $b - a$. For a range of width 2ε the exponent is $\lambda^2 \varepsilon^2/2$.

3.5 Composition bounds

Basic adaptive composition of $(\varepsilon_i, \delta_i)$-DP mechanisms gives $(\sum_i \varepsilon_i, \sum_i \delta_i)$-DP. The advanced composition theorem of Dwork–Rothblum–Vadhan (2010), with the improved mean term of Dwork–Rothblum (2016, Theorem 1.1), states that k adaptive (ε, δ_0)-DP mechanisms together satisfy

$$\left(\varepsilon\sqrt{2k\log(1/\eta)} + \frac{k\varepsilon(e^{\varepsilon} - 1)}{2},\ k\delta_0 + \eta\right)\text{-DP} \qquad (0 < \eta < 1). \tag{3.5}$$

The individual mechanism's slack, often written δ', is renamed δ_0 here, and the new parameter chosen when the theorem is applied, often written δ, is renamed η. Both are additive terms in the final guarantee, but only η is a free choice. The condition $\eta > 0$ is needed for $\log(1/\eta)$ to be finite; at $\eta = 0$, basic composition is still available.

3.6 Where the improved mean bound comes from

For pure DP, $R = p/q$ satisfies $a = e^{-\varepsilon} \le R \le b = e^{\varepsilon}$ Q-almost surely, and $\mathbb{E}_Q[R] = \int p \, \mathrm{d}\nu = 1$. The mean privacy loss is a Kullback–Leibler (KL) divergence:

$$\mathbb{E}_P[L] = \int p \log \frac{p}{q} \, \mathrm{d}\nu = \int qR \log R \, \mathrm{d}\nu = \mathbb{E}_Q[R \log R] = D_{\mathrm{KL}}(P \| Q).$$

Why consider $r \log r$? It is exactly the function inside the expectation we need to bound. It is convex, since $(r \log r)'' = 1/r > 0$, so on $[a, b]$ its graph lies below the chord joining its values at a and b:

$$R \log R \le \frac{b - R}{b - a} \, a \log a + \frac{R - a}{b - a} \, b \log b \qquad Q\text{-a.s.}$$

The right-hand side is linear in R, so taking $\mathbb{E}_Q$ (justified by the almost-sure inequality) simply replaces R by $\mathbb{E}_Q[R] = 1$:

$$D_{\mathrm{KL}}(P \| Q) \le \frac{b - 1}{b - a} \, a \log a + \frac{1 - a}{b - a} \, b \log b.$$

The algebra. Substitute $\log a = -\varepsilon$, $\log b = \varepsilon$ and $ab = 1$. The numerator is $\varepsilon\big[-(b-1)a + (1-a)b\big] = \varepsilon(a + b - 2ab) = \varepsilon(a + b - 2)$. Writing $s = e^{\varepsilon/2}$, we have $a + b - 2 = (s - s^{-1})^2$ and $b - a = (s - s^{-1})(s + s^{-1})$, so

$$D_{\mathrm{KL}}(P \| Q) \le \varepsilon \, \frac{s - s^{-1}}{s + s^{-1}} = \varepsilon \tanh \frac{\varepsilon}{2} = \varepsilon \, \frac{e^{\varepsilon} - 1}{e^{\varepsilon} + 1} \le \frac{\varepsilon(e^{\varepsilon} - 1)}{2}. \tag{3.6}$$

The last step uses $e^{\varepsilon} + 1 \ge 2$. The looser right-hand side is the mean term in (3.5); the chord gives the sharper $\varepsilon \tanh(\varepsilon/2)$, which is at most $\varepsilon^2/2$ because $\tanh u \le u$ for $u \ge 0$.

3.7 Where the square-root term comes from

We prove (3.5) for $\delta_0 = 0$. Sample the transcript on dataset x, so all probabilities are under P. Let $\mathcal{F}_{i-1}$ be the σ-field of the first $i - 1$ outputs. Conditionally on the history, round i is an ε-DP mechanism, so its mean loss is a KL divergence bounded by (3.6):

$$\mu_i = \mathbb{E}[L_i \mid \mathcal{F}_{i-1}] \le \mu_0 = \frac{\varepsilon(e^{\varepsilon} - 1)}{2}.$$

Let $X_i = L_i - \mu_i$. Then $\mathbb{E}[X_i \mid \mathcal{F}_{i-1}] = 0$, and since $L_i \in [-\varepsilon, \varepsilon]$, X_i lies in $[-\varepsilon - \mu_i, \varepsilon - \mu_i]$, an interval of width 2ε that is fixed once the history is. Hoeffding's lemma, applied conditionally, gives

$$\mathbb{E}[e^{\lambda X_i} \mid \mathcal{F}_{i-1}] \le e^{\lambda^2 \varepsilon^2/2}.$$

The tower step. Let $S_j = \sum_{i \le j} X_i$. Because S_{j-1} is fixed by $\mathcal{F}_{j-1}$,

$$\mathbb{E}[e^{\lambda S_j}] = \mathbb{E}[e^{\lambda S_{j-1}}\, \mathbb{E}[e^{\lambda X_j} \mid \mathcal{F}_{j-1}]] \le e^{\lambda^2 \varepsilon^2/2}\, \mathbb{E}[e^{\lambda S_{j-1}}],$$

and by induction $\mathbb{E}[e^{\lambda S_k}] \le e^{k\lambda^2 \varepsilon^2/2}$. Chernoff's bound (3.4) then gives, for every $\lambda > 0$,

$$\Pr[S_k \ge t] \le \exp\left(-\lambda t + \frac{k\lambda^2 \varepsilon^2}{2}\right).$$

The exponent is a parabola in λ with its minimum where $-t + k\lambda\varepsilon^2 = 0$, that is at $\lambda = t/(k\varepsilon^2)$. There it equals $-t^2/(2k\varepsilon^2)$, so $\Pr[S_k \ge t] \le \exp[-t^2/(2k\varepsilon^2)]$. Choosing $t = \varepsilon\sqrt{2k \log(1/\eta)}$ makes this at most η.

Finally $L_{1:k} = S_k + \sum_i \mu_i \le S_k + k\mu_0$, so

$$\Pr_P[L_{1:k} > \varepsilon_*] \le \eta, \qquad \varepsilon_* = \varepsilon\sqrt{2k \log(1/\eta)} + k\mu_0.$$

Lemma 3.1 turns this tail bound into $P(S) \le e^{\varepsilon_*} Q(S) + \eta$. The hypotheses are symmetric in x and x', so the same argument with the roles exchanged gives the other direction. This proves (3.5) when $\delta_0 = 0$.

When $\delta_0 > 0$. One needs an extra reduction, which we cite rather than prove. An (ε, δ_0)-indistinguishable pair can be written as $P = (1-\delta_0)P' + \delta_0 P''$ and $Q = (1-\delta_0)Q' + \delta_0 Q''$, where P' and Q' are ε-indistinguishable (DRV10; Kairouz–Oh–Viswanath 2015 give the canonical form). Running the argument above on the primed parts and charging each round's exceptional part to the slack costs at most $k\delta_0$ in total. It is **not** valid to assume that (ε, δ_0)-DP bounds $\Pr[L > \varepsilon]$ by δ_0: the example in Section 3.3 shows that the probability can be far larger.

4 CDP: mean-concentrated differential privacy (Dwork–Rothblum)

4.1 Motivation and definition

Advanced composition used two facts about L beyond its worst-case value: its mean, and how fast deviations above the mean become unlikely. Dwork–Rothblum make these two quantities the *definition* of privacy.

A real random variable X is τ-subgaussian if $\mathbb{E}[e^{\lambda(X - \mathbb{E}X)}] \le \exp(\lambda^2 \tau^2/2)$ for every real λ. A mechanism is (μ, τ)-CDP when, for every ordered neighbouring pair, the privacy loss under P satisfies

$$\mathbb{E}_P[L] \le \mu, \qquad \mathbb{E}_P[e^{\lambda(L-\mathbb{E}_P L)}] \le e^{\lambda^2\tau^2/2} \quad \text{for every } \lambda \in \mathbb{R}. \tag{4.1}$$

This version of CDP is also called mean-concentrated DP (mCDP), to distinguish it from zCDP. Unlike pure DP, it does not require a finite worst-case bound on the loss.

4.2 CDP to approximate DP

Apply Chernoff's bound (3.4) to $X = L - \mathbb{E}L$ with the subgaussian bound:

$$\Pr[L - \mathbb{E}L \ge t] \le \inf_{\lambda>0} \exp\left(-\lambda t + \frac{\lambda^2\tau^2}{2}\right).$$

The exponent is minimized where $-t + \lambda\tau^2 = 0$, at $\lambda = t/\tau^2$, where it equals $-t^2/(2\tau^2)$:

$$\Pr[L - \mathbb{E}L \ge t] \le e^{-t^2/(2\tau^2)}.$$

Choose $t = \tau\sqrt{2\log(1/\eta)}$ to make the right-hand side η. Since $\mathbb{E}L \le \mu$, the event $\{L > \mu + t\}$ is contained in $\{L - \mathbb{E}L \ge t\}$, so

$$\Pr[L > \mu + \tau\sqrt{2\log(1/\eta)}] \le \eta.$$

Lemma 3.1, applied to both ordered pairs, gives for each $0 < \eta < 1$

$$(\mu, \tau)\text{-CDP} \Longrightarrow (\mu + \tau\sqrt{2\log(1/\eta)},\ \eta)\text{-DP}. \tag{4.2}$$

4.3 Adaptive CDP composition: what the simple argument proves

Suppose that for every history, round i is (μ_i, τ_i)-CDP. Let $m_i = \mathbb{E}[L_i \mid \mathcal{F}_{i-1}]$, the conditional mean, which is at most μ_i but depends on the history. For $\lambda > 0$, replacing m_i by the larger constant μ_i can only lower the exponent, so

$$\mathbb{E}[e^{\lambda(L_i-\mu_i)} \mid \mathcal{F}_{i-1}] \le \mathbb{E}[e^{\lambda(L_i-m_i)} \mid \mathcal{F}_{i-1}] \le e^{\lambda^2\tau_i^2/2}.$$

The tower step of Section 3.7 then gives

$$\mathbb{E}\left[e^{\lambda(L_{1:k}-\sum_i \mu_i)}\right] \le \exp\left(\tfrac{\lambda^2}{2}\textstyle\sum_i \tau_i^2\right) \qquad (\lambda > 0). \tag{4.3}$$

This is a **one-sided, uncentred** bound, and it is all that the conversion (4.2) uses: rerunning that derivation with (4.3) gives

$$k \text{ adaptive rounds} \Longrightarrow \left(\textstyle\sum_i \mu_i + \sqrt{\sum_i \tau_i^2}\,\sqrt{2\log(1/\eta)},\ \eta\right)\text{-DP}. \tag{4.4}$$

For identical rounds this is $k\mu + \tau\sqrt{2k\log(1/\eta)}$, the mean-plus-square-root structure of advanced composition.

What this argument does not prove. The statement that the composition is again $(\sum_i \mu_i, \sqrt{\sum_i \tau_i^2})$-CDP, in the sense of (4.1), does not follow from multiplying conditional MGFs. Definition (4.1) centres $L_{1:k}$ at its unconditional mean $\sum_i \mathbb{E}[m_i]$, but the conditional bounds centre each L_i at the random m_i; the difference $\sum_i (m_i - \mathbb{E}m_i)$ is not controlled by the per-round hypotheses. For $\lambda < 0$, moreover, replacing m_i by μ_i moves the inequality the wrong way. Dwork–Rothblum (2016) state a composition theorem for CDP; Bun–Steinke (2016) discuss the technical difficulties of the mCDP definition. This note relies only on (4.3)–(4.4), which it proves.

4.4 Pure DP and Gaussian noise as CDP examples

Pure DP. Under ε-DP, $L \in [-\varepsilon, \varepsilon]$ P-a.s., so $L - \mathbb{E}L$ lies in an interval of width 2ε and Hoeffding's lemma makes it ε-subgaussian. With the mean bound (3.6),

$$\varepsilon\text{-DP} \Longrightarrow \left(\frac{\varepsilon(e^\varepsilon - 1)}{2},\ \varepsilon\right)\text{-CDP}.$$

Gaussian noise. Take a scalar Gaussian mechanism $M(x) = f(x) + N(0, \sigma^2)$ with $|f(x) - f(x')| \le \Delta$, and let $d = f(x) - f(x')$. A sample under P is $Y = f(x) + Z$ with $Z \sim N(0, \sigma^2)$, so $Y - f(x) = Z$ and $Y - f(x') = Z + d$. The normalizing constants cancel, and

$$L(Y) = \frac{(Y - f(x'))^2 - (Y - f(x))^2}{2\sigma^2} = \frac{(Z + d)^2 - Z^2}{2\sigma^2} = \frac{d^2}{2\sigma^2} + \frac{dZ}{\sigma^2}. \tag{4.5}$$

So under P the loss is Gaussian with mean $d^2/(2\sigma^2)$ and variance $d^2\sigma^2/\sigma^4 = d^2/\sigma^2$. A centred Gaussian with variance v has MGF $\mathbb{E}e^{\lambda X} = e^{\lambda^2 v/2}$, so the centred loss is $(|d|/\sigma)$-subgaussian, with equality. Bounding $|d| \le \Delta$,

$$M \text{ is } \left(\frac{\Delta^2}{2\sigma^2},\ \frac{\Delta}{\sigma}\right)\text{-CDP}. \tag{4.6}$$

The loss in (4.5) is unbounded, since Z is, so the Gaussian mechanism is **not** ε-DP for any finite ε. This is the case that concentrated notions were built for.

5 zCDP: the Rényi-divergence formulation (Bun–Steinke)

5.1 Why introduce Rényi divergence?

CDP controls the whole MGF of the centred privacy loss, and Section 4 showed that the centring is what makes composition awkward. Bun–Steinke instead bound the **Rényi divergence** at every order $\alpha > 1$. This keeps the useful tail bounds and makes adaptive composition clean.

For $P \ll Q$, the Rényi divergence of order $\alpha > 1$ is

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\int\left(\frac{\mathrm{d}P}{\mathrm{d}Q}\right)^{\alpha}\mathrm{d}Q, \tag{5.1}$$

and $D_\alpha(P\|Q) = +\infty$ when $P \not\ll Q$. Using $R = e^L$, the integral is $\mathbb{E}_Q[e^{\alpha L}]$. Since $\mathrm{d}P = R\,\mathrm{d}Q$,

$$\mathbb{E}_Q[e^{\alpha L}] = \mathbb{E}_Q\big[R\cdot R^{\alpha-1}\big] = \mathbb{E}_P\big[R^{\alpha-1}\big] = \mathbb{E}_P\big[e^{(\alpha-1)L}\big],$$

and therefore

$$D_\alpha(P\|Q) = \frac{1}{\alpha-1}\log\mathbb{E}_P\big[e^{(\alpha-1)L}\big]. \tag{5.2}$$

This is the exact bridge between the privacy loss and Rényi divergence: D_α is the cumulant generating function of L under P, evaluated at $\lambda = \alpha - 1$ and divided by λ. In the limit $\alpha \downarrow 1$, $D_\alpha \to D_{\mathrm{KL}}$ under the usual regularity assumptions, and as $\alpha \to \infty$, D_α tends to the max-divergence $\operatorname{ess\,sup}_P L$.

5.2 Definition of zCDP and its MGF form

A mechanism is (ξ,ρ)-zCDP when, for all neighbouring ordered pairs and all $\alpha > 1$,

$$D_\alpha(P\|Q) \le \xi + \rho\alpha, \qquad \xi,\rho \ge 0. \tag{5.3}$$

When $\xi = 0$ one simply says ρ-zCDP.

Insert (5.3) into (5.2) and set $\lambda = \alpha - 1 > 0$, so that $\alpha = 1 + \lambda$:

$$\log\mathbb{E}_P[e^{\lambda L}] \le \lambda(\xi + \rho(1+\lambda)) = \lambda(\xi+\rho) + \rho\lambda^2,$$

or equivalently

$$\mathbb{E}_P\big[e^{\lambda(L-\xi-\rho)}\big] \le e^{\rho\lambda^2}, \qquad \lambda > 0. \tag{5.4}$$

Compared with a Gaussian MGF $e^{\lambda^2\tau^2/2}$, the variance proxy is $\tau^2 = 2\rho$ and the location is $\xi + \rho$. This is a one-sided, uncentred statement, the same shape as (4.3); it is not the mCDP definition, which bounds a centred MGF for every real λ.

5.3 zCDP to approximate DP

For $t > 0$ and $\lambda > 0$, Chernoff's bound (3.4) with (5.4) gives

$$\Pr[L > \xi + \rho + t] \le \exp(-\lambda t + \rho\lambda^2).$$

For $\rho > 0$ the exponent is minimized where $-t + 2\rho\lambda = 0$, at $\lambda = t/(2\rho)$. There it equals $-t^2/(2\rho) + t^2/(4\rho) = -t^2/(4\rho)$, so

$$\Pr[L > \xi + \rho + t] \le e^{-t^2/(4\rho)}.$$

Choose $t = 2\sqrt{\rho \log(1/\eta)}$; then $t^2/(4\rho) = \log(1/\eta)$ and the tail probability is at most η. Lemma 3.1, in both directions, proves

$$(\xi, \rho)\text{-zCDP} \Longrightarrow (\xi + \rho + 2\sqrt{\rho \log(1/\eta)},\ \eta)\text{-DP}, \qquad 0 < \eta < 1. \tag{5.5}$$

When $\rho = 0$, the Rényi bound does not depend on α; letting $\alpha \to \infty$ bounds the max-divergence by ξ, which is pure ξ-DP. Formula (5.5) remains a valid, weaker statement in that case.

The conversion (5.5) optimizes only λ in a Chernoff bound, so it is not tight. Sharper conversions exist (Bun–Steinke 2016; Canonne–Kamath–Steinke 2020), and for the Gaussian mechanism the exact answer is computed in Section 7.

5.4 Adaptive composition of zCDP

Define each L_i from the conditional transcript densities, as in Section 2, and suppose every round is (ξ_i, ρ_i)-zCDP for every history. Fix $\alpha > 1$. By (5.2) and (5.3), for every history,

$$\mathbb{E}[e^{(\alpha-1)L_i} \mid \mathcal{F}_{i-1}] \le \exp((\alpha - 1)(\xi_i + \rho_i \alpha)).$$

The tower step of Section 3.7, with $\lambda = \alpha - 1$ and the uncentred L_i in place of X_i, bounds the MGF of $L_{1:k} = \sum_i L_i$ by the product of these bounds. Taking logarithms and dividing by $\alpha - 1$ gives

$$D_\alpha(P_{1:k} \| Q_{1:k}) \le \sum_i \xi_i + \alpha \sum_i \rho_i.$$

No centring appears, which is why the argument that failed for mCDP works here. Since α was arbitrary,

$$\text{adaptive composition is } \Big(\textstyle\sum_i \xi_i,\ \sum_i \rho_i\Big)\text{-zCDP}. \tag{5.6}$$

Combined with (5.5), k mechanisms that are each ρ-zCDP give $(k\rho + 2\sqrt{k\rho \log(1/\eta)},\ \eta)$-DP. Both zCDP parameters add linearly, while the final approximate-DP ε has a square-root term.

5.5 Post-processing

Rényi divergence satisfies the data-processing inequality: for any (possibly randomized) map g, $D_\alpha(g(P)\|g(Q)) \le D_\alpha(P\|Q)$ (cited, not proved here; see van Erven–Harremoës 2014). Hence zCDP is closed under post-processing. Bun–Steinke (2016) point out that mCDP, as defined by (4.1), is not.

5.6 The Gaussian mechanism has an exact zCDP parameter

Return to the Gaussian privacy loss (4.5), with $d = f(x) - f(x')$:

$$L \sim N\left(\frac{d^2}{2\sigma^2},\ \frac{d^2}{\sigma^2}\right) \quad \text{under } P.$$

For a Gaussian $N(m, v)$, $\log \mathbb{E}e^{\lambda L} = \lambda m + \lambda^2 v/2$, so here $\log \mathbb{E}_P[e^{\lambda L}] = \lambda d^2/(2\sigma^2) + \lambda^2 d^2/(2\sigma^2) = \lambda(1+\lambda)\, d^2/(2\sigma^2)$. Divide by $\lambda = \alpha - 1$ and use $1 + \lambda = \alpha$:

$$D_\alpha(P\|Q) = \frac{\alpha d^2}{2\sigma^2} \le \alpha\, \frac{\Delta^2}{2\sigma^2}.$$

This holds with equality when $|d| = \Delta$, so the Gaussian mechanism satisfies

$$\boxed{\rho = \frac{\Delta^2}{2\sigma^2}\text{-zCDP},} \tag{5.7}$$

and no smaller ρ works. This agrees with the CDP parameters (4.6), $\mu = \rho$ and $\tau = \sqrt{2\rho}$. For a vector query with independent noise in each coordinate, the same calculation holds with Δ the ℓ_2-sensitivity.

5.7 Pure DP implies zCDP

Under ε-DP, $L \in [-\varepsilon, \varepsilon]$ almost surely, and Hoeffding's lemma applied to $L - \mathbb{E}_P L$ gives

$$\log \mathbb{E}_P[e^{\lambda L}] \le \lambda\, \mathbb{E}_P L + \frac{\lambda^2 \varepsilon^2}{2}.$$

The chord bound (3.6) gives $\mathbb{E}_P L \le \varepsilon \tanh(\varepsilon/2) \le \varepsilon^2/2$, so

$$\log \mathbb{E}_P[e^{\lambda L}] \le \frac{\varepsilon^2}{2}(\lambda + \lambda^2) = \frac{\varepsilon^2}{2}\,\lambda(1+\lambda).$$

Divide by $\lambda = \alpha - 1$ and substitute $1 + \lambda = \alpha$: $D_\alpha(P\|Q) \le \alpha\varepsilon^2/2$. Therefore

$$\varepsilon\text{-DP} \Longrightarrow \frac{\varepsilon^2}{2}\text{-zCDP}. \tag{5.8}$$

This route explains why a quadratic parameter appears when a pure-DP guarantee is converted to a concentrated one. It needs the sharper mean bound $\varepsilon \tanh(\varepsilon/2)$; the looser $\varepsilon(e^\varepsilon - 1)/2$ exceeds $\varepsilon^2/2$ and would not give (5.8).

5.8 From mCDP to zCDP

The two notions are related in one direction, which follows from (5.2). Under (μ, τ)-CDP, for $\lambda > 0$,

$$\log \mathbb{E}_P[e^{\lambda L}] = \lambda\, \mathbb{E}_P L + \log \mathbb{E}_P[e^{\lambda(L - \mathbb{E}_P L)}] \le \lambda\mu + \frac{\lambda^2 \tau^2}{2}.$$

Divide by $\lambda = \alpha - 1$: $D_\alpha(P\|Q) \le \mu + (\alpha-1)\tau^2/2 = (\mu - \tau^2/2) + \alpha\,\tau^2/2$. If $\mu < \tau^2/2$ the constant is negative and can be dropped. Hence

$$(\mu, \tau)\text{-CDP} \Longrightarrow \Big(\max\{0,\, \mu - \tfrac{\tau^2}{2}\},\ \tfrac{\tau^2}{2}\Big)\text{-zCDP}. \tag{5.9}$$

For the Gaussian mechanism, $\mu = \tau^2/2 = \rho$, and (5.9) returns exactly (5.7). The converse does not follow from this note's arguments: zCDP controls L only through (5.4), a one-sided bound for $\lambda > 0$, while mCDP needs a centred bound for every real λ.

6 Comparing the guarantees without confusing the symbols

Notion	What it bounds	Adaptive composition	Conversion to (ε, η)-DP	Post-processing
Pure ε-DP	Every event-probability ratio; $\lvert L\rvert \le \varepsilon$ a.s.	$\sum_i \varepsilon_i$, pure DP	is already DP	closed
(ε, δ)-DP	$H_\varepsilon \le \delta$ in both directions	basic: $(\sum \varepsilon_i, \sum \delta_i)$; advanced: (3.5)	is already DP	closed
(μ, τ)-CDP (mCDP)	Mean loss $\le \mu$; centred MGF $\le e^{\lambda^2\tau^2/2}$	one-sided bound (4.3), giving (4.4)	(4.2)	not closed (BS16)
(ξ, ρ)-zCDP	Every Rényi divergence $D_\alpha \le \xi + \rho\alpha$	$(\sum \xi_i, \sum \rho_i)$-zCDP, (5.6)	(5.5)	closed

Bridges derived above: ε-DP $\Rightarrow (\varepsilon(e^\varepsilon - 1)/2, \varepsilon)$-CDP; ε-DP $\Rightarrow \varepsilon^2/2$-zCDP (5.8); (μ, τ)-CDP $\Rightarrow$ zCDP (5.9); Gaussian: $\rho = \Delta^2/(2\sigma^2)$, $\mu = \rho$, $\tau = \sqrt{2\rho}$.

Parameter warning. Sources use δ both for the additive slack of an approximate-DP guarantee **and** for a failure probability chosen freely during a conversion or in advanced composition. The two play different roles in the derivations but end up added together in an approximate-DP

statement. This note uses η for a newly chosen tail or conversion parameter, and keeps δ or δ_0 for the slack of an existing mechanism's guarantee.

7 Worked comparison: repeating a Gaussian query

A scalar query has sensitivity $\Delta = 1$ and is answered with Gaussian noise of standard deviation $\sigma = 5$, $k = 100$ times (adaptively if you like, provided every round has sensitivity at most 1 given the history). The target is $\delta_{\text{total}} = 10^{-5}$.

Through zCDP. Each round has $\rho = 1/(2 \cdot 25) = 0.02$. Composition (5.6) gives $\rho_{\text{total}} = 100 \times 0.02 = 2$. With $\eta = 10^{-5}$, so that $\log(1/\eta) = 11.513$, the conversion (5.5) gives

$$\varepsilon_{\text{total}} = 2 + 2\sqrt{2 \times 11.513} = 2 + 9.597 \approx 11.60, \qquad \delta_{\text{total}} = 10^{-5}.$$

Through mCDP. Each round is $(0.02, 0.2)$-CDP by (4.6), so (4.4) with $\sum \mu_i = 2$ and $\sqrt{\sum \tau_i^2} = \sqrt{100 \times 0.04} = 2$ gives $2 + 2\sqrt{2 \times 11.513} \approx 11.60$, the same number. That is no accident: for Gaussian noise the two definitions carry the same information.

The exact answer. For a pair of Gaussians, (3.3) can be evaluated in closed form. Write $\mu_G = |d|/\sigma$. By (4.5), under P the loss is $L = \mu_G^2/2 + \mu_G W$ with $W \sim N(0,1)$. Under Q the same calculation with $Y = f(x') + Z$ gives $L = -\mu_G^2/2 + \mu_G W$. The optimal event in (3.2) is $\{L > \varepsilon\}$, so $H_\varepsilon = P(L > \varepsilon) - e^\varepsilon Q(L > \varepsilon)$, that is,

$$\delta(\varepsilon) = \Phi\left(\frac{\mu_G}{2} - \frac{\varepsilon}{\mu_G}\right) - e^\varepsilon \, \Phi\left(-\frac{\mu_G}{2} - \frac{\varepsilon}{\mu_G}\right), \tag{7.1}$$

where Φ is the standard normal distribution function (Balle–Wang 2018). One round has $\mu_G = \Delta/\sigma = 0.2$. For 100 non-adaptive rounds with $|d| = \Delta$, the total loss is a sum of 100 independent copies of (4.5), which is again Gaussian, of mean $100 \times 0.02 = 2$ and variance $100 \times 0.04 = 4$. That is the loss of a single Gaussian with $\mu_G = \sqrt{k}\,\Delta/\sigma = 2$. (That adaptive rounds are no worse is the composition theorem for Gaussian DP, Dong–Roth–Su 2022; cited, not derived.) Solving (7.1) for $\delta = 10^{-5}$ with $\mu_G = 2$ gives $\varepsilon \approx 10.00$.

Through per-round (ε, δ)-DP. Formula (7.1) with $\mu_G = 0.2$ gives each round's ε_0 for a chosen δ_0:

Route	Per-round guarantee	$\varepsilon_{\text{total}}$ at $\delta_{\text{total}} = 10^{-5}$
Basic composition	$(0.932,\ 10^{-7})$-DP	93.2
Advanced composition (3.5), best split of δ	$(0.938,\ 8.5 \times 10^{-8})$-DP, $\eta = 1.5 \times 10^{-6}$	121.6
zCDP or mCDP, then convert	$\rho = 0.02$	11.60
Exact, (7.1)	$\mu_G = 0.2$	10.00

Advanced composition loses to basic composition here because $\varepsilon_0 \approx 0.94$ is not small: its mean term $k\varepsilon_0(e^{\varepsilon_0} - 1)/2 \approx 72.8$ alone is most of the basic total. The concentrated notions avoid the problem because they never replace the Gaussian loss by a worst case.

All of these are **guarantees**, upper bounds that hold for every pair of neighbours; none says the realized privacy loss equals 11.60. Increasing the noise lowers ρ quadratically: doubling σ divides each ρ by four. And never equate an ε-DP parameter numerically with a zCDP ρ: they measure different objects.

8 Limitations and exact scope of the derivations

The steps for pure DP, the hockey-stick formula and the tail lemma, the KL chord bound, the subgaussian tails, the Rényi–MGF bridge, the Gaussian privacy loss, zCDP composition, and the CDP-to-zCDP bridge are derived in this note. Four results are cited rather than proved: Hoeffding's lemma; the decomposition of an (ε, δ_0)-indistinguishable pair needed for advanced composition with $\delta_0 > 0$ (which is **not** implied by the false shortcut $\Pr[L > \varepsilon] \leq \delta$); the data-processing inequality for Rényi divergence; and the adaptive composition theorem for Gaussian DP used for the exact figure. For mCDP, this note proves only the one-sided composition bound (4.3)–(4.4), not closure of the definition (4.1) under composition.

CDP/mCDP and zCDP are related, by (5.9), but not interchangeable. An arbitrary approximate-DP mechanism need not have finite Rényi divergence at any order, and hence need not satisfy zCDP: any mechanism with $\delta > 0$ that puts mass where its neighbour puts none has $D_\alpha = \infty$. All comparisons assume one fixed adjacency convention (add/remove versus replace-one) and one privacy unit; sensitivities, and with them every number in Section 7, change with the convention.

9 Primary sources and reading order

1. C. Dwork, G. Rothblum, S. Vadhan (2010), *Boosting and Differential Privacy.* Original advanced-composition method and the $O(\varepsilon^2)$ mean-loss result. https://privacytools.seas.harvard.edu/sites/g/files/omnuum6656/files/privacytools/files/05670947.pdf
2. C. Dwork, G. Rothblum (2016), *Concentrated Differential Privacy.* Theorem 1.1, the original (μ, τ)-CDP, concentration and Gaussian analysis, improved expected-loss bound. https://arxiv.org/abs/1603.01887
3. M. Bun, T. Steinke (2016), *Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds.* Definition of (ξ, ρ)-zCDP, Rényi-divergence conversions, additive composition, the Gaussian mechanism, and the relation to mCDP. https://arxiv.org/abs/1605.02065
4. P. Kairouz, S. Oh, P. Viswanath (2015), *The Composition Theorem for Differential Privacy.* Optimal composition beyond the standard advanced bound, and the canonical form of an (ε, δ) pair. https://proceedings.mlr.press/v37/kairouz15.pdf
5. B. Balle, Y.-X. Wang (2018), *Improving the Gaussian Mechanism for Differential Privacy: Analytical Calibration and Optimal Denoising.* The exact privacy profile (7.1). https://arxiv.org/abs/1805.06530
6. J. Dong, A. Roth, W. Su (2022), *Gaussian Differential Privacy.* Composition of Gaussian mechanisms under adaptivity. https://arxiv.org/abs/1905.02383
7. C. Canonne, G. Kamath, T. Steinke (2020), *The Discrete Gaussian for Differential Privacy.* A sharper zCDP-to-DP conversion than (5.5). https://arxiv.org/abs/2004.00010

8. T. van Erven, P. Harremoës (2014), *Rényi Divergence and Kullback–Leibler Divergence*. Properties of D_α, including data processing. https://arxiv.org/abs/1206.2459